

CabinGuard-ADI: Bounded-Evidence Multimodal Detection of Acute Driver Incapacitation with an Authenticated Minimum-Risk Response

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Abstract—Driver monitoring systems detect drowsiness and distraction, not the sudden loss of driving capacity caused by cardiac syncope or a convulsive seizure. CabinGuard-ADI separates these two events from normal driving using four contactless channels: camera pulse rhythm, eyelid and head pose, a headrest accelerometer, and seat and steering-wheel sensors. The evidence of every channel is bounded so that no single sensor can trigger a vehicle response; a minimum-risk manoeuvre needs two physical sensors that agree for 2 seconds. The pulse-rhythm and motion branches are trained on public PhysioNet recordings, including 22.2 hours of road driving, ten annotated seizures, ventricular arrhythmia onsets and 10.4 hours of driving accelerometry, and are evaluated across held-out records and subjects. On out-of-fold composite episodes the fused detector started no manoeuvre in 22.2 hours of real driving, a 95 % upper bound of 0.14 per hour, against 3.92 false starts per hour for the pulse branch alone, and detected 79/80 syncope and 30/30 seizure episodes with median latencies of 12.0 and 13.2 seconds from onset. The manoeuvre is commanded through authenticated CAN frames and supervised by a gateway that rejects forged, replayed and implausible commands and completes the stop if the detector fails. A signed and encrypted 76-byte medical extension reaches the emergency centre, while nearby vehicles are warned without the diagnosis. The eye, head and seat channels still use modelled likelihoods, and a motion branch trained on injected seizure motion detected few recorded seizure mimics; the Phase 2 bench replaces both with measured signals.

Index Terms—Driver monitoring, driver incapacitation, syncope, epileptic seizure, multimodal fusion, remote photoplethysmography, minimum risk manoeuvre, SecOC, eCall

I. INTRODUCTION AND MOTIVATION

Sudden illness of the driver is a small but persistent cause of fatal crashes: in Finland and in the Swiss canton of Vaud it caused 1.5 % and 3.4 % of all traffic deaths, and probable cardiac arrest alone about 2 % of driver deaths [1]. Since July 2022 new European vehicles must carry drowsiness and distraction warning systems, and the same regulation requires those systems to work without biometric information [2]. Neither function addresses a driver who loses consciousness or motor control within seconds.

Two events dominate that scenario. Cardiac syncope is a transient loss of consciousness from cerebral hypoperfusion caused by an arrhythmia [3]; the driver becomes flaccid, the head drops and the hands leave the wheel. A convulsive

seizure begins with a tonic phase of sustained contraction and continues with clonic, rhythmic jerking [4]. The two look different only on average: myoclonic jerks occurred in 90 % of syncopes induced in healthy volunteers [5], so a motion sensor alone confuses them, while a camera alone cannot tell closed eyes from a blinded lens. An unnecessary manoeuvre on a motorway is itself a hazard.

The contributions are a fusion rule under which no single sensor can start a manoeuvre, with the bound derived from the decision threshold; pulse-rhythm and motion detectors evaluated on public real recordings, including real road driving; a gateway that accepts only authenticated, plausible commands and fails safe; and an emergency message that gives the etiology to the emergency centre alone.

II. RELATED WORK

Camera-based pulse measurement projects the skin colour signal onto a plane orthogonal to the skin tone (POS) [6]; in a moving car, narrow-band near-infrared video reduces the effect of changing daylight [7]. Real-road driver ECG comes from the driver-stress study of Healey and Picard [8], and eye closure is commonly measured with the eye aspect ratio [9].

Seizure detection outside hospitals relies mostly on wearables: a wrist accelerometer detected 35 of 39 generalized tonic-clonic seizures with 0.2 false alarms per day [10]. Clonic discharges last about 0.2 s with silent periods of exponentially increasing duration [11], and heart rate rises during most partial seizures [12]; none of these studies measures motion through a vehicle seat.

On the vehicle side, UN Regulation No. 157 defines a minimum risk manoeuvre that an automated lane keeping system performs when a transition demand is not answered [13]. In-vehicle networks without authentication let a compromised node command brakes [14], cameras can be blinded with commodity lasers [15], and AUTOSAR secure onboard communication (SecOC) adds truncated message authentication codes and freshness values to CAN frames [16].

III. PROBLEM DEFINITION

Every 100 ms the system assigns the driver to a class $C \in \{N, S, Z\}$ for normal driving, cardiac syncope and convulsive seizure, and decides whether to start a minimum risk manoeuvre (MRM). It must meet five requirements: R1, few false MRM starts per hour of normal driving; R2, detection

within seconds of onset; R3, no single sensor able to start an MRM; R4, safe behaviour under sensor, processing and communication failure and under tampered or unauthorized messages; R5, only the minimum necessary data leaving the vehicle. A false start first produces a 3 s alert that a conscious driver cancels, so R1 counts alerts as well as braking.

IV. SYSTEM ARCHITECTURE

The sensing layer has three physical sensors, shown on the left of Fig. 1. A 30 fps camera with 850 nm illumination on the steering column yields the pulse beat times, the eye aspect ratio e and the head pitch θ . An MPU6050 accelerometer in the headrest is sampled at 100 Hz. A force-sensing resistor in the seatback, a second one in the wheel rim, two capacitive touch pads and the steering torque τ form the seat and wheel sensor. An ESP32 node streams these signals in 24-byte frames with a sequence number and CRC to the edge host, which runs detection, the watchdog and the MRM state machine and commands the vehicle only through one authenticated CAN message to a gateway that owns the actuators. Every input carries a timestamp, and data older than 300 ms is treated as missing.

V. DETECTION METHODOLOGY

A. Features

The camera front end detects pulse peaks in the POS signal [6]; in the public recordings used for training the beats are ECG R-peaks instead. Over a 10 s window the pulse-rhythm vector $\mathbf{c}$ holds the rate from the median inter-beat interval, its difference from the driver's own running baseline, the coefficient of variation of the intervals, and the fraction a of the window not covered by intervals shorter than 2 s. Ventricular fibrillation, flutter and asystole produce no pulse, so they raise a towards one. Eye opening is the eye aspect ratio of Soukupová and Čech [9], computed from six eyelid landmarks.

$$e = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|} \quad (1)$$

In Equation (1), p_1 and p_4 are the eye corners and p_2, p_3, p_5, p_6 the upper and lower lid points in image coordinates. Clonic motion is characterised by the share of acceleration power in the 2–6 Hz band, computed over a 2 s window from the sum $S(f)$ of the three per-axis Welch spectra [17].

$$r = \frac{\int_2^6 S(f) df}{\int_{0.5}^{20} S(f) df} \quad (2)$$

In Equation (2), r is the spectral energy ratio and f the frequency in hertz. A ratio alone is noisy when the body is still, so the motion vector $\mathbf{m}$ adds the 2–6 Hz RMS acceleration, the peak frequency, the total RMS, the largest autocorrelation at lags of 0.15–0.5 s, which captures the repetition of jerks, and the peak-to-mean ratio of the band spectrum. Postural slump ψ is one minus the mean normalized seatback pressure over 1 s, and g is the grip state: active, disengaged, or clenched when the rim force exceeds a threshold.

B. Branch Evidence

Four branches turn features into evidence. The cardiac branch is a multinomial logistic regression on $\mathbf{c}$ and the motion branch gradient-boosted trees [18] with histogram splits [19] on $\mathbf{m}$, both trained with class weights that balance the classes. The vision branch on (e, θ) and the posture branch on (ψ, g) use expert-set Gaussian and categorical class densities, since no public recording provides them for incapacitated drivers. A discriminative model's posterior becomes a likelihood ratio once its training prior is divided out, the scaled-likelihood step of hybrid connectionist recognisers [20].

$$\lambda_b(C) = \ln \frac{P_b(C | \mathbf{x}_b)}{\pi_b(C)} - \ln \frac{P_b(N | \mathbf{x}_b)}{\pi_b(N)} \quad (3)$$

In Equation (3), $\lambda_b(C)$ is the log-likelihood ratio of class C against normal driving from branch b , $\mathbf{x}_b$ the branch input and π_b the prior the model was trained under, uniform here because of the balanced weights. The motion branch only separates clonic motion from everything else, so it gives $\lambda_{\text{mot}}(S) = 0$.

C. Bounded-Evidence Fusion

Each branch's evidence for an event is limited to a bound c by shifting both event classes together, which keeps the branch's preference between syncope and seizure.

$$\tilde{\lambda}_b(C) = \max\left(-c, \lambda_b(C) - \max\left(0, \max_{C' \neq N} \lambda_b(C') - c\right)\right) \quad (4)$$

In Equation (4), $\tilde{\lambda}_b(C)$ is the bounded evidence for $C \neq N$ and C' ranges over the two event classes. Treating the branches as conditionally independent given the class, the product rule for combining classifiers [21] gives the fused posterior.

$$P(C | \mathbf{x}) \propto \pi(C) \exp\left(\sum_{b \in \mathcal{A}} \tilde{\lambda}_b(C)\right) \quad (5)$$

In Equation (5), $\mathcal{A}$ is the set of branches whose sensors are available, $\bar{\lambda}_b$ the average of $\tilde{\lambda}_b$ over the last $W = 2$ s, and π the operational prior with $\pi(N) = 0.8$ and 0.1 for each event. With the decision threshold $\Gamma = 0.85$ and the prior log-odds $l_0 = \ln(0.1/0.8)$, the bound is chosen so that one branch cannot reach Γ while two agreeing branches can.

$$l_0 + c < \ln \frac{\Gamma}{1 - \Gamma} < l_0 + 2c \quad (6)$$

Equation (6) holds for $1.91 < c < 3.81$, and $c = 3$ is used. A missing branch simply leaves the sum in Equation (5), so sensor loss never needs a separate model.

D. Decision Rule

The watchdog sets the flag F that starts the MRM when one event class keeps $P(C | \mathbf{x}) \geq \Gamma$ for $T_{\text{ver}} = 2$ s and at least two physical sensors give $\tilde{\lambda}_b(C) \geq 1$. The camera feeds two branches but counts as one sensor, so a blinded lens reporting closed eyes and no pulse cannot decide alone.

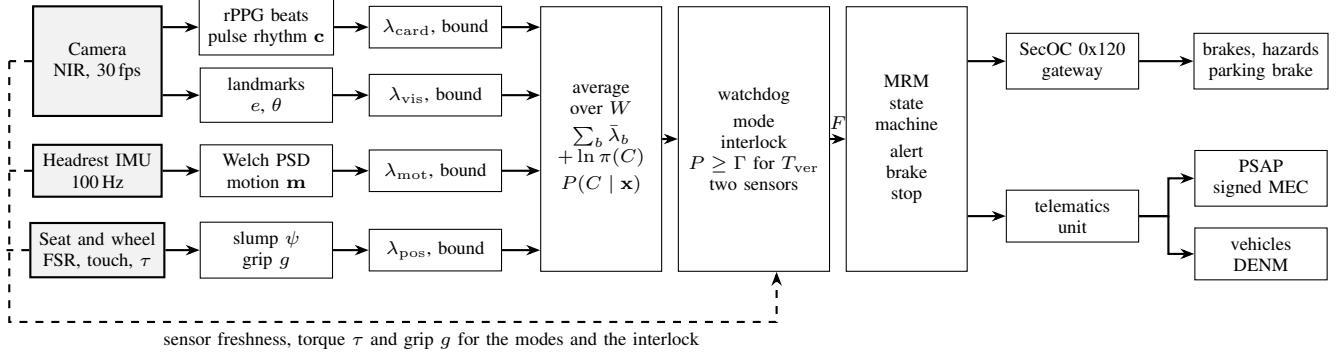


Fig. 1. CabinGuard-ADI signal flow from the three sensors to the vehicle actuators and the two emergency receivers.

VI. SAFETY RESPONSE AND FAULT HANDLING

Once F is set, the state machine follows the transition-demand sequence of UN Regulation No. 157 [13] with design values of our own: a 3 s audible and haptic alert that the driver cancels by braking or by more than 4 Nm of steering torque, unless the torque coincides with clonic motion or a clenched grip; hazard lights and escalation; braking at -3.2 m/s^2 from 10 s after the alert with a lateral move to the shoulder; and at standstill the parking brake and door unlock. The emergency call is placed when the alert goes unanswered, not after the stop.

The watchdog derives the mode from sensor freshness. Losing the camera removes the cardiac and vision branches (degraded mode 1), losing the accelerometer removes the motion branch (degraded mode 2), and with fewer than two sensors left no MRM may start and the driver is warned instead (degraded mode 3). When the camera reports no pulse or closed eyes while the wheel shows an upright driver steering with more than 1 Nm, the readings contradict each other, so the camera is distrusted for 10 s and the event is logged.

The gateway trusts no command. A frame must pass authentication and a plausibility gate that allows only the phase order of the state machine and decelerations between 0 and -4 m/s^2 . The command doubles as an alive signal: after 300 ms without a valid frame the gateway flags an ECU fault and, if the driver is still in control, actuates nothing; if a manoeuvre is already under way it completes the stop, because a driver judged incapacitated cannot take over. The host itself goes silent after three consecutive 100 ms deadline overruns. The emergency message is retried with 1, 2 and 4 s backoff on the cellular bearer, then sent by SMS, and otherwise kept and retried every 30 s; the manoeuvre never waits for it.

VII. CYBERSECURITY AND THREAT MODEL

The protected assets are the vehicle actuators, the driver's physiological data, the integrity of the emergency message and the driver's identity and location. The sensor link inside the cabin is protected against corruption by its CRC but not against an attacker with physical access, which is outside the scope; the CAN bus, the cellular network and the radio channel are untrusted, and the gateway is the trusted boundary in front

TABLE I
THREATS, CONTROLS AND THE FAULT-MATRIX CASE THAT EXERCISES EACH

Threat	Control	Test case
Forged command	24-bit AES-CMAC, SecOC profile 1	unauthorized key
Replayed command	64-bit freshness counter, 8 bits sent	replayed frames
Tampered frame	MAC check, alive timeout	tampered frames
Valid MAC, unsafe value	phase order, -4 m/s^2 limit	implausible command
Camera blinding or spoofing	cross-sensor interlock, two-sensor rule	optical blinding
Sensor loss or stale data	freshness, degraded modes	IMU, seat stale
ECU hang	fail-silent host, gateway takeover	two processing cases
Network outage	retry, SMS fallback, queue	two bearer cases
Message forgery, eavesdropping	ECDSA P-256 signature, ECIES to PSAP	unit tests

of the actuators. Table I lists the threats and the control that answers each.

The 0x120 command follows SecOC profile 1, AES-128-CMAC [22] truncated to 24 bits over the data identifier, payload and a 64-bit freshness value of which the low 8 bits are sent [16]. Its payload, the CAN database file and the decoder are tested to agree bit for bit. The status frame that other nodes can read carries sensor health only, never a biometric value.

The 76-byte medical extension container (MEC) has a 12-byte header (version, etiology, confidence, detection latency, sensor mask, heart rate, sequence number, timestamp) and a 64-byte ECDSA P-256 signature from a pseudonymous key, the curve and signature used by ITS security certificates [23]. It is encrypted to the emergency centre's public key with ephemeral ECDH, HKDF and AES-GCM and fits as optional additional data next to the 140-byte eCall minimum set of data [24], which already carries the location. Nearby vehicles receive a decentralized environmental notification with cause code 93, human problem, and sub-cause 0, unavailable [25], so the etiology never leaves the emergency channel. The system stores no video and no waveform; only 10 s feature buffers

TABLE II
PUBLIC RECORDINGS USED FOR TRAINING AND EVALUATION

Database	Content	Role
Driver stress [8]	ECG, 17 road drives	normal, 22.2 h
Partial epilepsy [12]	ECG, 10 seizures	seizure, normal
Malignant ventricular ectopy [27]	ECG, 22 records	syncope, normal
CU tachyarrhythmia [28]	ECG, 35 records	syncope, normal
MIT-BIH	ECG, 48 records	syncope, normal
arrhythmia [29]		
Walk, climb, drive [30]	100 Hz accel., 32 adults	motion, 10.4 h driving

exist, in memory.

VIII. EVALUATION METHODOLOGY

Table II lists the public recordings, all from PhysioNet [26]. Driving-study R-peaks come from the WFDB XQRS detector. Windows in VT are labelled cardiac and keep their beats; windows in fibrillation, flutter or asystole are labelled cardiac and lose their beats, because no pulse reaches the face. Atrial fibrillation, bigeminy and other rhythms that do not incapacitate count as normal, as hard negatives.

Branches are evaluated with leave-one-record-out validation for the cardiac branch (130 records) and leave-one-subject-out validation for the motion branch, using the hip accelerometer as the closest available proxy for a seat-mounted sensor. No public recording measures seizure motion in a seat, so motion positives are real driving windows with an added jerk train that follows the clonic dynamics of [11]: 0.2 s discharges with silent periods starting at 0.05–0.15 s and growing to about 1 s over 40 jerks, at amplitudes of 0.02 to 0.5 g. Only amplitudes of 0.1 g and above are used as training positives; all are scored.

The fused detector is evaluated on composite episodes, which is the conditional-independence assumption of Equation (5) applied to the test data. A normal episode pairs a whole held-out drive with held-out driving accelerometry. A syncope episode takes 80 ventricular arrhythmia onsets with 30 s of normal rhythm before them; loss of consciousness, which drives the vision and posture channels, follows after 6–12 s, an assumption. A seizure episode takes each of the ten seizures, a 5–15 s tonic phase and then injected clonic motion at 0.1, 0.2 and 0.5 g. Vision and posture channels are generated with everyday confounders: blinks, 1–3 s glances down every 45 s, forward leans every 120 s and one-handed steering every 60 s. Every model is applied out of fold. Two further checks use recordings downloaded outside the main pipeline: seizure mimics performed by healthy volunteers with a wrist accelerometer at 16 Hz [31], and everyday activities from a chest sensor [32] and a waist-worn phone [33]. The deployed motion branch is applied unchanged after resampling to 100 Hz, and, as a reference, the same features are trained on the mimic dataset’s own training split. Separately, the full software pipeline runs 14 failure and attack cases with synthetic sensors, 10 seeds each.

TABLE III
BRANCH RESULTS UNDER HELD-OUT RECORDS AND HELD-OUT SUBJECTS

Cardiac branch	Gaussian NB	Logistic	Boosted trees
AUC, VT/VF vs rest	0.928	0.955	0.936
AUC, seizure vs rest	0.825	0.846	0.790
Seizure records hit, of 7	7	5	6
Motion branch	Gaussian NB	Logistic	Boosted trees
AUC, driving vs clonic	0.861	0.818	0.857
Alarms per driving hour	7.0	9.2	4.8
Sensitivity at 0.05 g	0.17	0.34	0.07
Sensitivity at 0.1 g	0.74	0.54	0.91
Sensitivity at 0.2 g	1.00	0.90	1.00

TABLE IV
FUSED DETECTOR ON OUT-OF-FOLD COMPOSITE EPISODES

Branches	False/h	Syncope	Seizure	Latency, s
Cardiac only	3.92	47/80	15/30	7.5, 30.5
Motion only	0.00	0/80	27/30	–, 15.0
Vision only	0.00	80/80	30/30	12.0, 13.2
Posture only	1.13	79/80	30/30	14.0, 6.0
Cardiac, motion	0.14	0/80	23/30	–, 17.0
All four	0.00	79/80	30/30	12.0, 13.2
Camera lost	0.18	0/80	30/30	–, 14.0
IMU lost	0.14	79/80	27/30	12.0, 11.0

IX. RESULTS

Table III gives the branch results. The pulse rhythm separates ventricular arrhythmia well and seizures less well, which matches ictal heart-rate rises of 3 to 88 bpm in this database [12]. The fixed spectral-ratio rule $r > 0.65$ fires 34.3 times per hour of real driving, so road vibration alone defeats it, and its higher sensitivity to weak injected jerks (0.46 against 0.07 at 0.05 g) comes only from that; the learned motion branch reduces the rate to 4.8 per hour, still too many for a standalone trigger.

The deployed models are the logistic cardiac branch and the boosted-tree motion branch, each the best of the three on its own evaluation. Table IV shows what fusion adds. Only the cardiac and motion rows are scored on real signals; the vision and posture channels are generated from the same expert densities that score them, so their near-perfect rows are circular and every row that includes them inherits part of that optimism. On real signals alone, the cardiac branch raises 3.92 false starts per hour, and adding the motion branch cuts this to 0.14 but cannot confirm syncope, which the motion branch says nothing about. All four branches started no manoeuvre in 22.2 hours of real driving, which puts the 95 % upper bound of the rate at 0.14 per hour, and detected 79/80 syncope and 30/30 seizure episodes with 90th-percentile latencies of 14.1 and 19.0 s from onset. Losing the camera keeps seizure detection but, by design, confirms no syncope, since only the seat would still support it.

The averaging window W was searched over 0, 2, 4 and 8 s on these same episodes. The fused detector started no manoeuvre at any of them, and the median syncope latency grew from 11.5 to 16.5 s; $W = 2$ s is the shortest window

with the highest syncope sensitivity, a choice made on the test episodes and therefore optimistic.

The recorded seizure mimics expose the limit of training on injected motion. The deployed branch fired on only 3 of 34 mimics, and on 1 of 104 walking, running and sawing recordings; the same features trained on the mimic dataset fired on 33 of 34 mimics and 0 of the others. The features carry the information, but the seizure sensitivities of Table IV hold only for motion that resembles the injection model. On everyday activities the branch alone raised 10 alarms in 10 min of cycling and 4 during walking, none in the other activities, and flagged at most 0.8% of phone windows, which leaves the two-sensor rule to absorb rhythmic movement.

All 140 of 140 fault-matrix runs reached the required outcome: no manoeuvre under sensor loss, stale data, optical blinding, or tampered, replayed, forged or implausible commands; a fail-silent gateway when the host hangs during normal driving and a completed stop when it hangs during a manoeuvre; and a delivered, verified MEC after cellular loss or a 60 s loss of every bearer. On one laptop core the detection cycle took 2.7 ms at the median and 4.6 ms at the 99th percentile; the slowest of all cycles took 65 ms, which the three-overflow rule tolerates.

X. CONCLUSION

Bounding each sensor's evidence and requiring two physical sensors made the detector both sensitive and quiet on real driving, and the vehicle moves only on commands that a fail-safe gateway authenticates. Three limits remain. The vision and posture likelihoods are expert-set and their channels modelled, so the fused results above are an upper estimate until those channels are measured; seizure motion is injected rather than measured in a seat, and the branch trained on it misses most recorded seizure mimics until it is retrained on such recordings; and the composite episodes assume that the branches are independent given the class. The same code already runs on an ESP32 sensor node, a camera and a CAN bus; Phase 2 volunteer sessions on that bench will measure the eye, head and seat likelihoods, the seizure-mimic amplitude at the headrest, camera pulse accuracy and timing on the target processor.

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